tarea_3_Guillermo_Montero

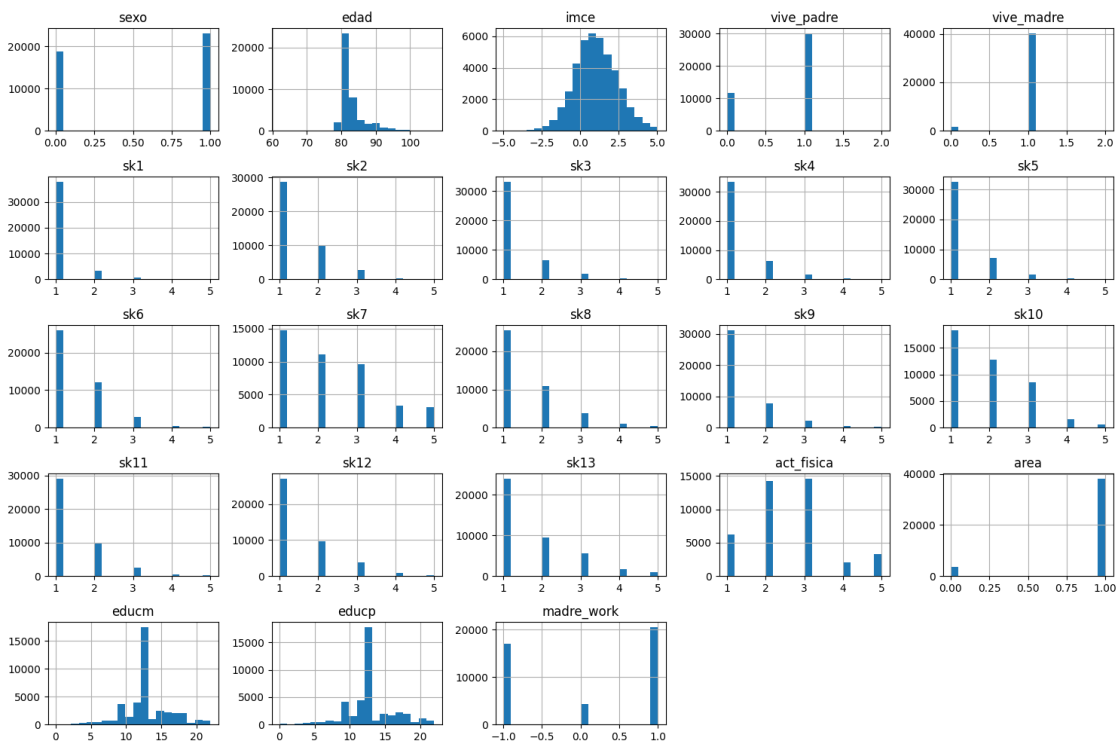
July 5, 2025

Pregunta 1: Cargue la base de datos y realice los ajustes necesarios para su uso (missing values, recodificar variables, etcetera). Identifique los tipos de datos que se encuentran en la base, realice estadisticas descriptivas sobre las variables importantes (Hint: Revisar la distribuciones, datos faltantes, outliers, etc.) y limpie las variables cuando sea necesario.

Primero es necesario caracterizar los datos y darles un poco de forma, aplicar algunas estadisticas descriptivas para entender la naturaleza de estos

```
               sexo          edad          imce    vive_padre    vive_madre  \
count  41854.000000  41854.000000  41854.000000  41854.000000  41854.000000
mean       0.552325     83.066039      1.014083      0.720051      0.965141
std        0.497261      3.987135      1.380418      0.449830      0.192328
min        0.000000     62.000000     -5.020000      0.000000      0.000000
25%        0.000000     81.000000      0.100000      0.000000      1.000000
50%        1.000000     82.000000      0.970000      1.000000      1.000000
75%        1.000000     84.000000      1.930000      1.000000      1.000000
max        1.000000    107.000000      5.040000      2.000000      2.000000

                sk1           sk2           sk3           sk4           sk5  \
count  41854.000000  41854.000000  41854.000000  41854.000000  41854.000000
mean       1.117958      1.397310      1.270129      1.263129      1.275935
std        0.401019      0.661753      0.594456      0.588868      0.575344
min        1.000000      1.000000      1.000000      1.000000      1.000000
25%        1.000000      1.000000      1.000000      1.000000      1.000000
50%        1.000000      1.000000      1.000000      1.000000      1.000000
75%        1.000000      2.000000      1.000000      1.000000      1.000000
max        5.000000      5.000000      5.000000      5.000000      5.000000

       …           sk9          sk10          sk11          sk12  \
count  …  41854.000000  41854.000000  41854.000000  41854.000000
mean   …      1.346920      1.887275      1.397620      1.511158
std    …      0.682243      0.959059      0.684203      0.811187
min    …      1.000000      1.000000      1.000000      1.000000
25%    …      1.000000      1.000000      1.000000      1.000000
50%    …      1.000000      2.000000      1.000000      1.000000
75%    …      2.000000      3.000000      2.000000      2.000000
max    …      5.000000      5.000000      5.000000      5.000000
```

```
                sk13    act_fisica          area         educm         educp  \
count  41854.000000  40419.000000  41854.000000  41303.000000  41854.000000
mean       1.715941      2.552611      0.911621     12.975280     12.897142
std        1.002515      1.069379      0.283848      3.373766      3.462212
min        1.000000      1.000000      0.000000      0.000000      0.000000
25%        1.000000      2.000000      1.000000     11.000000     11.000000
50%        1.000000      2.000000      1.000000     13.000000     13.000000
75%        2.000000      3.000000      1.000000     15.000000     14.000000
max        5.000000      5.000000      1.000000     22.000000     22.000000

         madre_work
count  41854.000000
mean       0.083457
std        0.943453
min       -1.000000
25%       -1.000000
50%        0.000000
75%        1.000000
max        1.000000

[8 rows x 23 columns]
```



0.0.1 analizar que valores estan presentes en cada parametro para serciorarse que sean valores dentro de los rangos permitidos

```
{'sexo'} [1 0]
{'edad'} [ 85  76  68  84  86  74  91  92  69  80  78  79  73  82  83  75  77
71
 107 106 105 104 103 102 101 100  99  98  97  96  95  94  93  90  89  81
  88  65  87  62]
{'imce'} [ 0.75        0.70999998  0.27000001  2.05        1.05        1.39
  2.75        1.53        1.96        1.76        0.          0.04
  1.9299999   1.65        0.52999997  1.14        2.53        1.75
  0.43000001  1.29        0.67000002 -1.45        3.4100001   2.98
 -0.12        1.77        2.03        1.1         2.0799999   2.7
 -0.02       -0.11       -0.40000001  1.9         0.60000002  0.61000001
 -2.3699999   2.52        0.5         1.9400001   1.25        1.41
 -0.55000001  0.63999999  0.31       -0.09       -1.15        0.66000003
  0.05        0.94999999  0.69999999  0.97000003  2.21        1.78
  2.9200001  -0.30000001  0.49000001  0.44999999  1.5700001   2.04
  0.16       -1.52        2.1600001  -0.03        4.1599998   0.55000001
  1.62        3.22        0.40000001 -0.25        3.1700001   0.80000001
  3.2         0.58999997 -0.60000002  0.94       -0.31        0.91000003
  0.23        0.17        1.64        2.6600001  -0.01        0.33000001
  2.1500001   2.1099999   0.38999999  2.0699999   0.23999999  2.4100001
 -1.35        1.34        0.95999998  3.96        1.45        2.26
 -3.3        -0.47        1.49        2.1300001   1.95        2.3499999
 -0.36000001 -1.37       -0.99000001  1.61        1.          3.47
  1.5         2.1800001   1.72        1.83        3.4400001   1.79
  3.46        4.3000002   0.63        1.6900001   0.86000001  0.92000002
  2.0999999  -0.34        0.08       -0.28999999  1.36       -0.54000002
 -1.05       -0.08        0.34999999  0.14        1.12       -0.58999997
  0.11        3.0699999  -0.07        3.5899999  -3.4000001   2.1700001
  0.47999999  2.27        1.88        0.28        1.63        1.92
  0.81        1.3        -0.76999998  1.23       -0.94999999  1.71
  0.15000001 -1.6900001   2.74        2.5799999   2.24        3.1500001
 -0.56       -1.3099999  -0.51999998  0.47       -0.38999999  0.46000001
  2.9100001   1.13        2.23        2.3599999   1.84        0.69
  0.07        3.6500001   2.3299999   2.9400001   1.4        -1.1799999
  0.83999997 -0.80000001  1.51       -0.18000001  1.8099999   3.54
  3.1300001   1.4299999   1.03       -0.41999999  0.36000001  2.3800001
 -1.12       -0.66000003  0.99000001 -1.14       -3.8         2.8699999
  0.68000001  4.2600002   0.93000001  2.4400001   2.3099999   2.8399999
  1.52        0.79000002  1.5599999   2.4300001  -1.16        1.1799999
 -1.65       -0.72000003  3.28        2.01        1.7         1.02
  3.3        -0.05        0.37        3.1099999  -0.68000001  2.99
  2.0599999   0.74000001  3.75        2.77       -0.81999999  3.23
 -1.49       -1.24        0.34        2.22        2.          2.51
 -0.77999997  0.06        1.54        2.46        1.16       -0.25999999
  1.55        0.57999998  0.64999998 -0.50999999  1.0700001   1.8200001
```

3.01	0.56	-0.23999999	0.31999999	0.98000002	0.73000002
-0.82999998	-0.49000001	-0.1	-0.73000002	1.0599999	2.6800001
-0.94	0.44	1.73	0.62	0.02	-0.44
-0.2	-0.04	-0.70999998	1.09	3.3599999	1.3099999
-0.98000002	3.04	-0.16	2.1199999	-0.93000001	-1.23
2.6700001	-0.15000001	2.29	0.41999999	-1.02	0.56999999
5.04	2.02	1.27	2.6300001	1.87	-0.13
-1.08	-0.92000002	1.91	1.15	3.3399999	-0.43000001
2.8099999	3.4300001	1.46	2.49	1.11	-1.75
2.28	0.76999998	-0.14	1.38	-2.51	3.53
-3.03	-0.75	0.82999998	1.47	-1.51	0.18000001
0.88	1.58	-0.19	3.27	-2.1400001	-1.28
-1.86	3.25	-0.38	0.75999999	1.01	2.5699999
3.6300001	-0.27000001	4.54	1.74	-1.89	-1.91
-3.25	2.8299999	0.25999999	-0.37	-0.62	0.51999998
2.72	-0.63	-1.6799999	-1.26	0.30000001	0.85000002
-3.6700001	1.85	2.5	-2.3199999	2.8199999	3.1199999
2.9300001	1.86	1.17	0.87	2.8499999	3.3499999
-0.81	-0.52999997	-0.85000002	-0.69999999	1.28	1.2
1.6799999	0.77999997	1.35	-0.5	0.89999998	-0.75999999
-0.88999999	-3.6500001	0.1	-1.67	2.3900001	1.21
-1.71	1.66	-0.22	2.48	-1.96	2.76
2.8900001	-0.91000003	3.7	2.6199999	1.98	3.03
2.1900001	3.3699999	-1.53	-1.9299999	2.54	0.72000003
-0.69	2.0899999	-1.11	-0.61000001	-0.41	-0.64999998
1.4400001	-1.7	3.5799999	3.8299999	-1.13	-1.54
4.	2.2	0.38	3.5999999	-1.33	2.6099999
2.47	0.54000002	-1.66	2.79	1.59	-1.42
1.48	-0.89999998	0.22	-0.31999999	-1.2	3.78
-1.03	-1.39	0.41	2.55	-1.0599999	2.73
2.3	1.22	1.8	1.67	1.04	-1.64
3.51	3.8699999	-3.1900001	3.05	1.08	2.4200001
-1.5	1.26	-1.04	2.9000001	-0.33000001	2.5899999
1.6	3.0599999	0.50999999	-2.96	-0.17	-1.4
3.48	2.45	0.03	2.6500001	0.88999999	0.20999999
2.3699999	-0.28	0.13	0.09	-0.06	-1.62
-0.67000002	3.3199999	-1.21	3.6700001	-0.74000001	2.25
4.6999998	-2.8599999	1.24	-1.29	3.26	3.24
-0.46000001	-1.3200001	-0.86000001	-1.47	2.3199999	-2.
-1.5599999	3.0999999	3.4000001	3.3900001	3.5699999	-1.74
1.37	-2.5899999	2.96	-1.34	-1.	2.6900001
1.99	1.89	-0.56999999	-0.23	4.25	3.1600001
3.02	-1.17	3.8	-1.98	-1.58	4.6399999
0.81999999	-0.88	-2.74	4.1300001	-2.28	1.3200001
2.95	3.3299999	-0.63999999	3.6600001	3.8900001	-0.44999999
0.12	1.97	0.01	-0.20999999	-1.87	3.1900001
0.2	2.8	-3.6800001	-2.3900001	-2.29	4.5700002
1.1900001	4.9000001	1.33	4.4000001	0.25	2.78

3.	-2.27	2.97	3.73	-1.61	-2.25
-0.83999997	3.6400001	3.45	-2.1099999	2.4000001	4.0799999
-1.9400001	-1.09	-2.3399999	2.5599999	4.4499998	4.4899998
-1.3	-2.0799999	-0.57999998	2.5999999	3.8099999	-1.46
2.3399999	-1.84	-0.97000003	-0.95999998	-1.1900001	4.8699999
1.42	3.8599999	3.3800001	4.2399998	3.71	4.3600001
-0.79000002	-1.0700001	4.48	-2.9200001	3.3099999	4.0599999
-0.47999999	3.76	-1.88	3.8399999	-1.6	-3.1099999
3.72	-2.6199999	-1.63	-2.26	3.9200001	4.3400002
-3.3599999	-2.0699999	3.9300001	3.4200001	3.52	4.77
-1.22	3.77	4.9899998	3.49	-2.6900001	2.8599999
-1.59	-1.8200001	0.19	-2.05	3.8499999	4.9499998
-1.8099999	2.6400001	-0.34999999	3.0899999	-2.79	3.1800001
-2.9000001	-1.8	3.79	3.74	4.6700001	-2.0999999
4.73	-2.3599999	4.4299998	4.5300002	-2.71	-1.79
4.21	-4.27	-2.04	4.1799998	-2.24	4.6500001
4.8099999	4.1100001	-1.83	2.1400001	-1.48	4.7199998
4.1399999	-2.9300001	2.8800001	-4.1300001	4.9099998	2.71
4.4400001	4.4099998	4.29	4.3299999	4.46	4.1999998
-1.5700001	-1.85	3.9400001	-2.4300001	-0.87	4.0700002
4.6599998	-1.38	4.04	-2.72	-2.03	-3.3199999
3.0799999	-4.4899998	-2.3099999	-2.8199999	4.0900002	-1.72
-2.48	4.3800001	-2.46	4.3499999	0.28999999	-2.01
-3.1300001	3.1400001	-4.8499999	4.3699999	3.6099999	-1.36
-1.77	-1.4400001	-2.8900001	-1.78	4.2199998	4.0500002
-1.27	3.6199999	-1.41	-2.1199999	-2.45	5.0100002
-1.25	-1.55	-2.76	-2.23	-2.0599999	-2.2
4.5799999	-1.1	4.02	-2.6400001	-2.75	-3.8199999
3.5	3.8800001	3.55	-2.21	4.2800002	3.21
-2.1800001	-1.01	3.98	4.1199999	4.5999999	-2.3299999
-2.0899999	3.29	4.8200002	-2.4400001	-2.02	4.6799998
4.98	-3.28	3.9000001	-1.99	4.3099999	4.4699998
-4.8400002	-1.95	-3.8099999	-1.97	-2.8	4.79
4.6199999	4.6900001	4.7800002	3.8199999	3.9100001	4.1900001
-3.02	-4.3600001	5.0300002	-1.73	3.99	4.27
-2.1300001	-2.9400001	4.23	4.5500002	4.8499999	-1.4299999
-1.76	-1.9	4.96	-3.23	-2.73	-2.1500001
3.6900001	-4.6599998	4.1500001	4.5599999	3.6800001	-2.8099999
-1.92	4.0100002	-2.1600001	-3.1700001	3.5599999	4.71
-2.5599999	4.1700001	4.0999999	4.3899999	-3.4300001	-3.55
4.3200002	-2.4100001	-3.71	4.9200001	4.5900002	-4.9400001
-2.55	-2.8699999	-4.4000001	-2.4200001	4.8000002	-3.05
3.95	-2.5999999	4.0300002	4.7399998	-4.4099998	-4.7600002
-2.97	4.8800001	-2.9100001	4.5	4.5100002	-3.8900001
-4.4299998	4.8600001	-3.21	-3.3900001	-2.8800001	-3.8599999
-2.1900001	-2.3499999	-2.77	-4.0900002	-4.6300001	-3.26
-2.8399999	-3.27	4.9699998	-4.3499999	-3.77	4.52
-3.1600001	4.4200001	-4.3099999	4.6100001	-3.1500001	-2.3800001

```
 -3.52       -2.3         3.97       -3.51       -2.52       -2.47
 -2.5699999  -4.7199998  -4.5300002  -2.6700001  -2.22       -3.0999999
 -2.53       -3.3800001  -3.74        4.6300001  -2.78       -2.8499999
 -2.99       -3.97       -2.5799999  -4.5100002  -4.6999998  -5.02
 -3.24       -2.4000001   4.8400002  -3.29       -4.2800002  -3.4400001
  4.7600002   4.8299999  -3.8800001  -3.4200001  -2.98       -2.1700001
 -2.5         5.         -2.6800001  -4.71       -3.01       -2.6500001
 -3.3699999  -2.6099999  -4.6900001  -4.4699998  -3.9000001  -4.6700001
 -2.54       -2.7        -3.45       -3.3499999  -4.04       -3.8499999
 -3.47       -4.4200001  -4.0100002  -2.8299999  -4.3000002   4.9400001
 -4.8600001  -3.22       -4.8200002  -4.1199999  -3.8399999  -4.6399999
 -4.73       -3.95        4.75       -3.75       -2.6600001  -4.02
 -3.8699999  -4.1399999  -4.3699999  -3.6400001  -4.1900001   5.02
  4.9299998  -2.49       -3.6600001  -4.5599999  -4.6199999  -4.2399998
 -3.3299999  -3.6199999  -4.23       -2.95       -3.         -3.0899999
 -3.53       -4.0599999  -4.9499998 ]
{'vive_padre'} [1 0 2]
{'vive_madre'} [1 0 2]
{'sk1'} [1 2 3 5 4]
{'sk2'} [1 2 3 4 5]
{'sk3'} [1 3 2 4 5]
{'sk4'} [2 1 3 4 5]
{'sk5'} [1 2 3 4 5]
{'sk6'} [2 1 3 5 4]
{'sk7'} [2 4 3 1 5]
{'sk8'} [1 2 4 3 5]
{'sk9'} [2 1 3 5 4]
{'sk10'} [2 1 3 4 5]
{'sk11'} [2 1 3 4 5]
{'sk12'} [3 1 2 5 4]
{'sk13'} [2 1 3 4 5]
{'act_fisica'} [nan  5.  2.  1.  4.  3.]
{'area'} [0 1]
{'educm'} [11.  8. 13. 16. 17. 20. 10.  6.  9. 12.  7.  4. 15. 19. 18.  0.  5.
3.
 nan 22. 14.  2.]
{'educp'} [11  8 13 12 15 19  9 17 18 20  7  2  5 16 10  0  4  3 22  6 14]
{'madre_work'} [-1  1  0]
```

0.0.2 También transformamos las edades de meses a años, junto con esto, agregamos una columna de los meses que actualmente tienen antes de cumplir otro año, también se invierte el signo de sk7

```
          edad  edad_años  edad_meses
0     7.083333          7           1
1     6.333333          6           4
2     5.666667          5           8
3     7.000000          7           0
```

```
4      7.166667          7           2
…         …        …          …
41849  6.500000          6           6
41850  6.583333          6           7
41851  6.500000          6           6
41852  6.500000          6           6
41853  6.500000          6           6

[41854 rows x 3 columns]

       sexo      edad  imce  vive_padre  vive_madre  sk1  sk2  sk3  sk4  sk5  \
1         0  6.333333  0.71           0           1    1    1    1    1    1
3         1  7.000000  2.05           1           1    1    1    1    1    1
4         0  7.166667  1.05           1           1    1    1    1    1    1
5         0  6.166667  1.39           1           1    1    2    1    1    1
6         1  7.583333  2.75           1           1    1    1    1    2    2
…     …        …   …          …          …  …  …  …  …  …
41849     0  6.500000  1.63           1           1    1    3    1    2    2
41850     1  6.583333  2.57           1           1    1    1    2    2    1
41851     0  6.500000  2.12           1           1    1    1    1    1    1
41852     1  6.500000 -0.43           1           1    1    1    1    1    2
41853     0  6.500000 -0.55           0           1    2    2    1    1    1

       …  sk11  sk12  sk13  act_fisica  area  educm  educp  madre_work  \
1      …     1     1     1         5.0     0    8.0      8           1
3      …     1     1     1         2.0     1   16.0     12          -1
4      …     1     1     1         1.0     1   17.0     15           0
5      …     1     1     1         4.0     0    8.0      8          -1
6      …     3     2     2         2.0     1   20.0     19           1
…    …   …   …   …          …   …    …    …          …
41849  …     2     1     1         2.0     1   13.0     13          -1
41850  …     2     1     4         3.0     1   18.0     19           0
41851  …     1     1     1         3.0     1   13.0      9           1
41852  …     1     1     2         2.0     1   17.0     15           1
41853  …     1     1     1         1.0     1   18.0     11           1

       edad_años  edad_meses
1              6           4
3              7           0
4              7           2
5              6           2
6              7           7
…           …          …
41849          6           6
41850          6           7
41851          6           6
41852          6           6
```

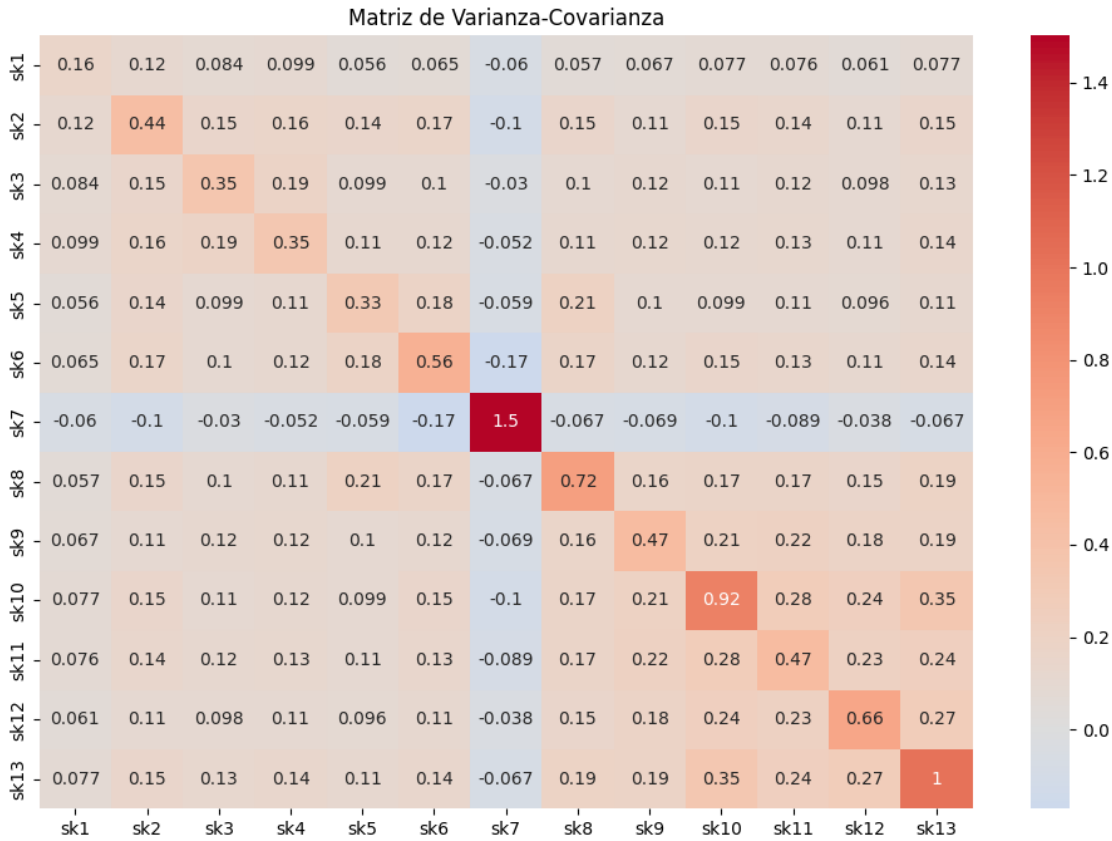
```
41853          6           6

[39898 rows x 25 columns]
```

Pregunta 2: Usando las variables sk1-sk13 realice un PCA. En particular, identifique los valores propios y determine el numero optimo de componentes. Luego estime y grafique la distribucion de los componentes. Ademas discuta la importancia relativa de las variables sobre cada uno de los componentes estimados. Que se puede concluir de este analisis?

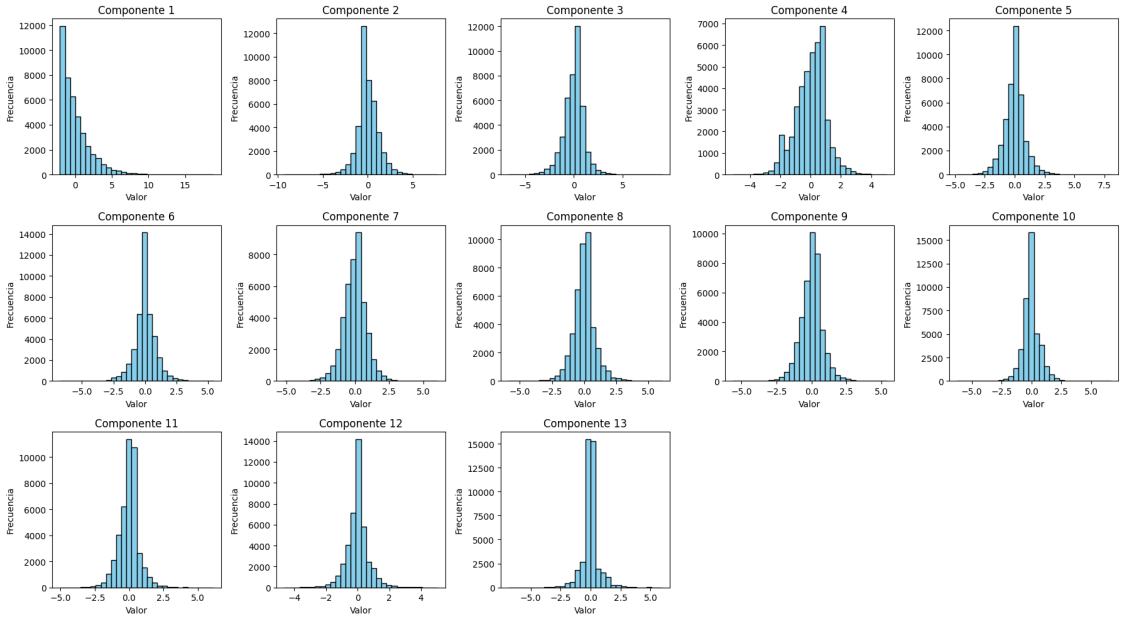
Matriz de Varianza-Covarianza:

	sk1	sk2	sk3	sk4	sk5	sk6	sk7 \
sk1	0.160816	0.121890	0.084232	0.099418	0.055879	0.065479	-0.059660
sk2	0.121890	0.437917	0.148950	0.163177	0.143776	0.167329	-0.100785
sk3	0.084232	0.148950	0.353378	0.185030	0.098853	0.103283	-0.029640
sk4	0.099418	0.163177	0.185030	0.346765	0.114236	0.119261	-0.052037
sk5	0.055879	0.143776	0.098853	0.114236	0.331021	0.181345	-0.059160
sk6	0.065479	0.167329	0.103283	0.119261	0.181345	0.557886	-0.171452
sk7	-0.059660	-0.100785	-0.029640	-0.052037	-0.059160	-0.171452	1.501665
sk8	0.056942	0.150658	0.102061	0.112041	0.206584	0.169575	-0.066551
sk9	0.066859	0.107903	0.115756	0.123011	0.101556	0.120640	-0.068758
sk10	0.076710	0.145020	0.105093	0.118927	0.099153	0.150613	-0.103642
sk11	0.075931	0.136644	0.117805	0.127016	0.109500	0.134608	-0.088950
sk12	0.060579	0.107231	0.097719	0.105240	0.095922	0.114836	-0.037975
sk13	0.076778	0.150207	0.126265	0.138541	0.110831	0.138658	-0.067342

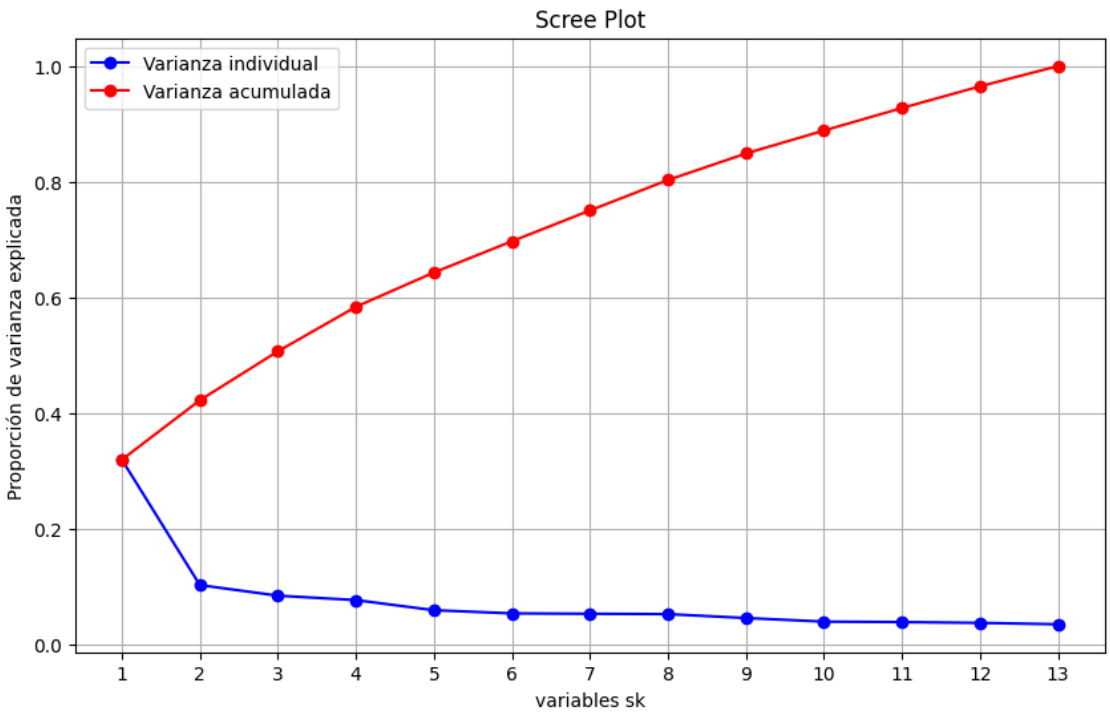
	sk8	sk9	sk10	sk11	sk12	sk13
sk1	0.056942	0.066859	0.076710	0.075931	0.060579	0.076778
sk2	0.150658	0.107903	0.145020	0.136644	0.107231	0.150207
sk3	0.102061	0.115756	0.105093	0.117805	0.097719	0.126265
sk4	0.112041	0.123011	0.118927	0.127016	0.105240	0.138541
sk5	0.206584	0.101556	0.099153	0.109500	0.095922	0.110831
sk6	0.169575	0.120640	0.150613	0.134608	0.114836	0.138658
sk7	-0.066551	-0.068758	-0.103642	-0.088950	-0.037975	-0.067342
sk8	0.716269	0.163766	0.170525	0.169764	0.153081	0.188281
sk9	0.163766	0.465456	0.206025	0.218397	0.175423	0.185925
sk10	0.170525	0.206025	0.919795	0.277375	0.238421	0.349313
sk11	0.169764	0.218397	0.277375	0.468134	0.225607	0.240110
sk12	0.153081	0.175423	0.238421	0.225607	0.658024	0.274440
sk13	0.188281	0.185925	0.349313	0.240110	0.274440	1.005037




```
[[ 0.99173088  1.37116719  0.58758255 …  0.19602155  0.13949011
   1.23485196]
 [-1.83422399 -0.572447   -0.50026172 …  0.27280703  0.07133031
  -0.05159715]
 [ 3.43462297 -0.99003682  1.65765512 … -0.98535343  0.35421771
  -1.0469    ]
 …
 [-1.52511168  0.48590559  0.65987299 …  0.31093164 -0.32274882
  -0.06124731]
 [-0.36621802 -0.12287925 -0.97598573 …  0.61291871  0.110111
  -0.32231078]
 [-0.57980857 -1.85479421 -0.0973969  …  0.23239956 -0.0200325
  -0.45371988]]
```

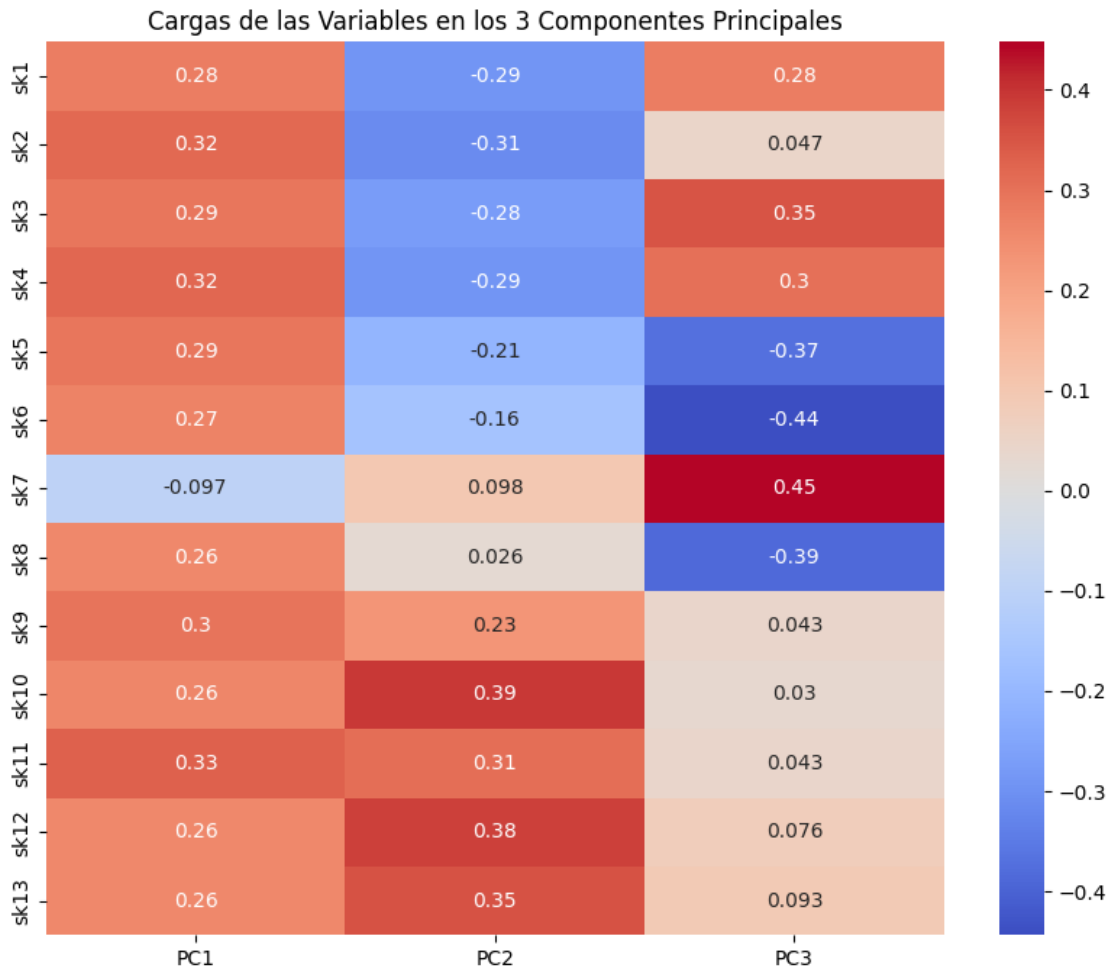


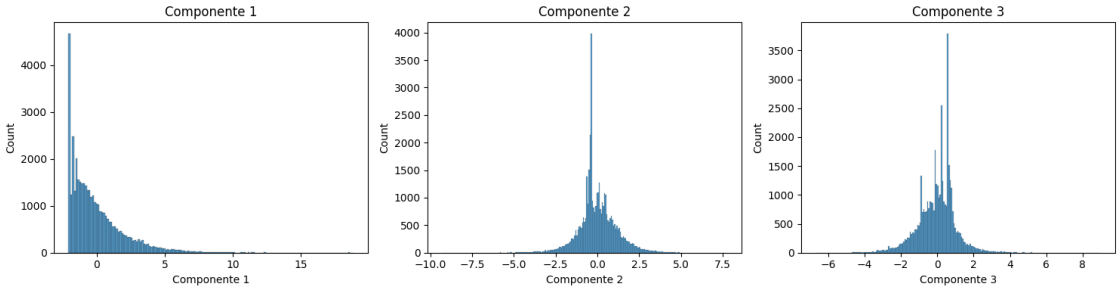

```
Valores propios:
[4.15436549 1.3365652  1.09910896 0.99951711 0.77210514 0.70114337
 0.69214386 0.68409742 0.59662788 0.5155363  0.50584496 0.4874235
 0.45583142]

Varianza explicada acumulada:
[0.31955894 0.42236919 0.50691402 0.58379811 0.64318939 0.69712221
 0.75036277 0.8029844  0.84887775 0.88853344 0.92744366 0.96493688
 1.        ]
```



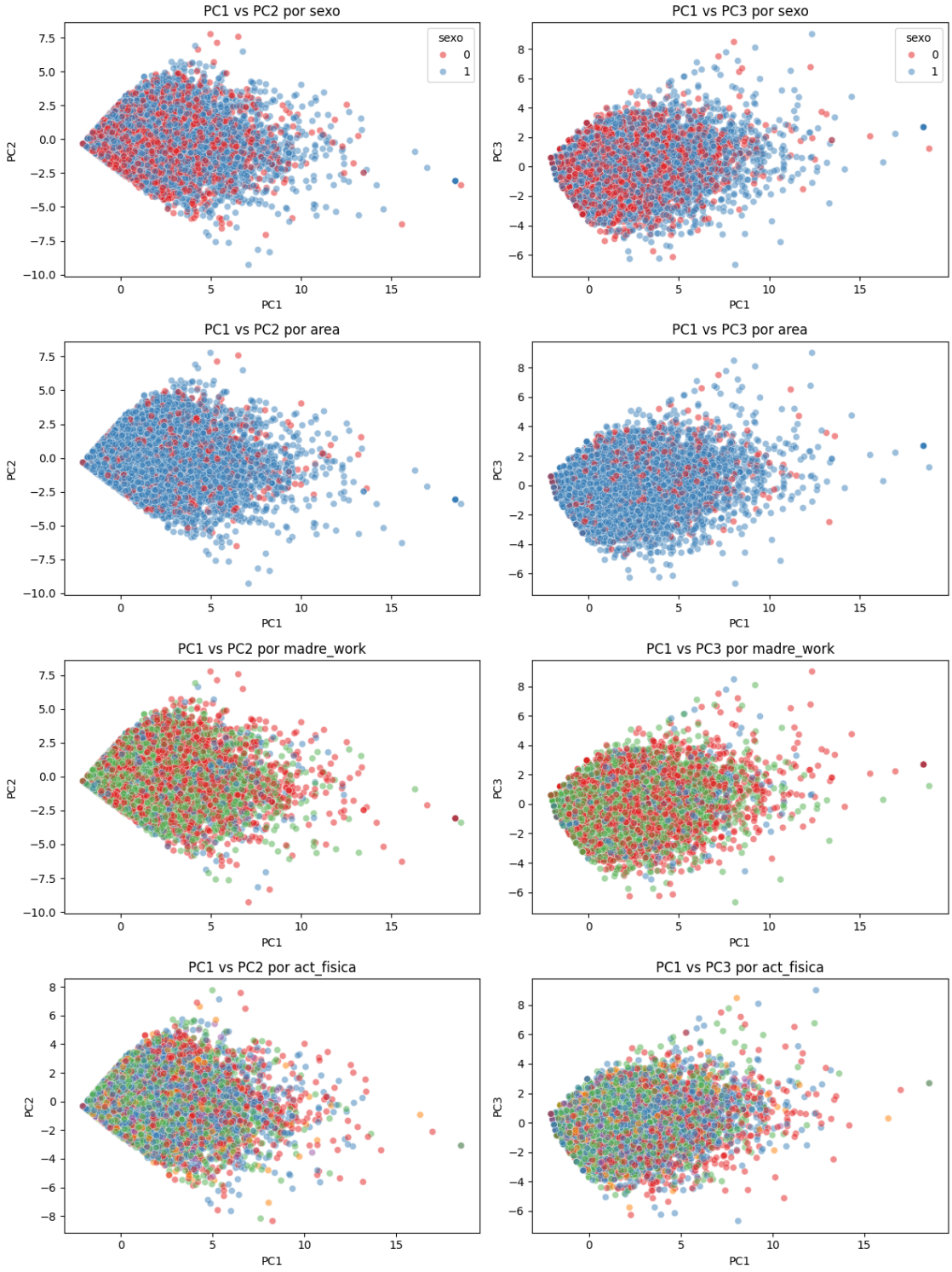
En vista de que los primeros 3 componentes tienen valores propios significativos, realizamos PCA con 3 componentes

```
Cargas de las variables en cada componente:
           PC1       PC2       PC3
sk1   0.279653 -0.292463  0.279524
sk2   0.316201 -0.313067  0.047264
sk3   0.293847 -0.275114  0.352083
sk4   0.320114 -0.290624  0.303434
sk5   0.294515 -0.208955 -0.372546
sk6   0.267577 -0.164967 -0.443692
sk7  -0.097080  0.097966  0.448011
sk8   0.259011  0.026300 -0.385174
sk9   0.296544  0.231483  0.042543
sk10  0.262194  0.392498  0.030369
sk11  0.330260  0.307261  0.042749
sk12  0.257829  0.384105  0.075601
sk13  0.256734  0.354336  0.092720
```





3. Con los resultados de la Pregunta 2, mantenga los primeros 3 componentes principales. Graficamente indique si existen diferencias significativas entre grupos usando las siguientes variables: sexo, area, madre_work y act_fisica. Que puede concluir de los resultados?

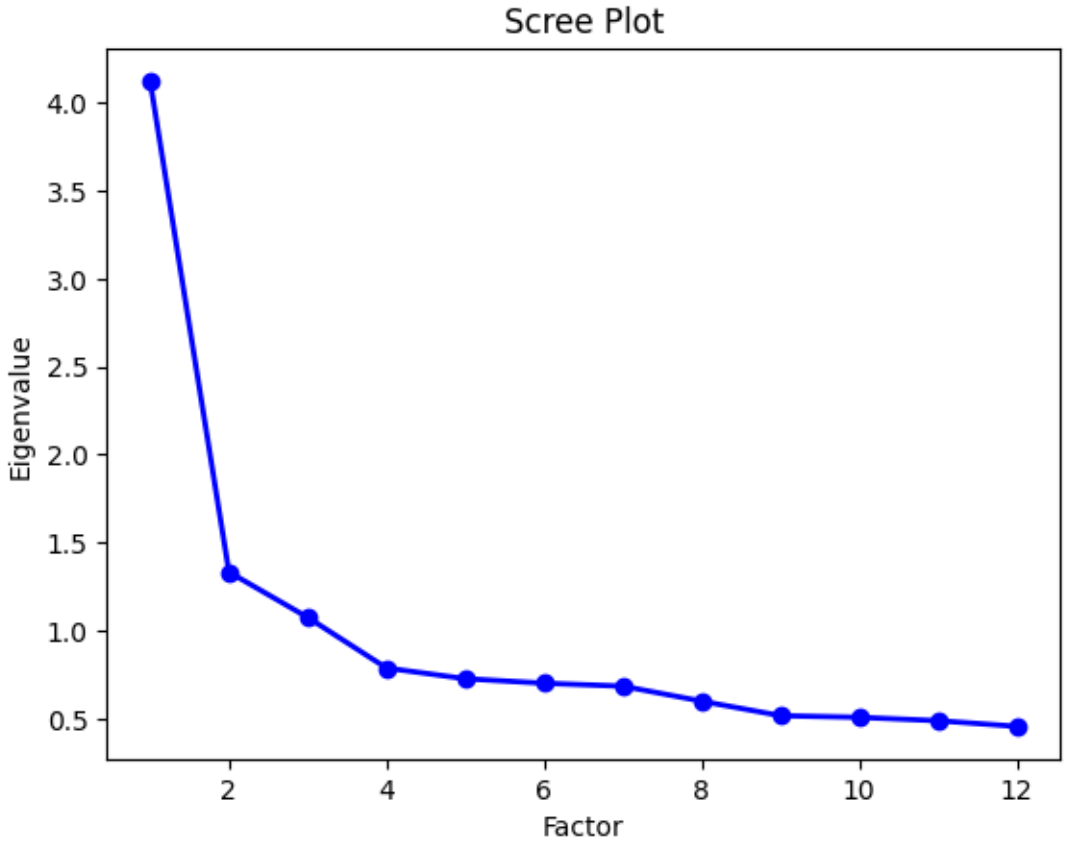


4. A partir del mismo set de variables sk1-sk13 realice un EFA. En particular determine el numero optimo de factores y las variables que se asocian a cada factor. Tambien

discuta si existen variables que no son informativas (Hint: para realizar un EFA, todas las variables deben estar representatadas en el mismo sentido logico. Si una carateristica es negativa debe ser invertida en la escala, de tal forma que todas las variables representen aspectos positivos).

```
d:\VSCODE ENVIROMENTS\proyectos universidad\data analysis y ML\LAB-
MAA\notebooks\.venv\Lib\site-packages\sklearn\utils\deprecation.py:132:
FutureWarning: 'force_all_finite' was renamed to 'ensure_all_finite' in 1.6 and
will be removed in 1.8.
  warnings.warn(
```

```
array([[0.023972, 0.612775, -0.039271],
       [0.007854, 0.524294, 0.187951],
       [0.016791, 0.651873, -0.021858],
       [-0.001211, 0.736657, -0.008124],
       [-0.028339, 0.017251, 0.823906],
       [0.105271, 0.103327, 0.409711],
       [0.236554, -0.064221, 0.441652],
       [0.505329, 0.087732, 0.050570],
       [0.619386, -0.033095, -0.030541],
       [0.697787, 0.038263, 0.008667],
       [0.577668, -0.032837, 0.005541],
       [0.537320, 0.015562, -0.012449]])
```



utilizando el criterio kaiser (eigenvalues > 1), se entiende que el numero optimo de factores es 3.

5. Con los resultados obtenidos en la Pregunta 4, proponga un CFA donde cada variable solo se asocia con un factor. Entregue un nombre a cada factor que representa el concepto comun entre todas las variables. Reporte la importancia de cada medida (variable) a cada factor e indique la correlacion entre factores.

```
d:\VSCODE ENVIROMENTS\proyectos universidad\data analysis y ML\LAB-
MAA\notebooks\.venv\Lib\site-packages\sklearn\utils\deprecation.py:132:
FutureWarning: 'force_all_finite' was renamed to 'ensure_all_finite' in 1.6 and
will be removed in 1.8.
  warnings.warn(
d:\VSCODE ENVIROMENTS\proyectos universidad\data analysis y ML\LAB-
MAA\notebooks\.venv\Lib\site-
packages\factor_analyzer\confirmatory_factor_analyzer.py:753: UserWarning: The
optimization routine failed to converge: ABNORMAL:
  warnings.warn(
```

```
 array([[1.000000, 0.139775, 0.145129],
        [0.139775, 1.000000, 0.156513],
        [0.145129, 0.156513, 1.000000]])
```

```
       Factor1   Factor2   Factor3
sk1   0.023972  0.612775 -0.039271
sk2   0.007854  0.524294  0.187951
sk3   0.016791  0.651873 -0.021858
sk4  -0.001211  0.736657 -0.008124
sk5  -0.028339  0.017251  0.823906
sk6   0.105271  0.103327  0.409711
sk8   0.236554 -0.064221  0.441652
sk9   0.505329  0.087732  0.050570
sk10  0.619386 -0.033095 -0.030541
sk11  0.697787  0.038263  0.008667
sk12  0.577668 -0.032837  0.005541
sk13  0.537320  0.015562 -0.012449

Variables asociadas a cada factor (carga > 0.3):
Factor1: ['sk9', 'sk10', 'sk11', 'sk12', 'sk13']
Factor2: ['sk1', 'sk2', 'sk3', 'sk4']
Factor3: ['sk5', 'sk6', 'sk8']
```

6. Finalmente, implemente un SEM completo usando la estructura propuesta en la Pregunta 5. En particular, estime un modelo donde los factores explican el nivel de actividad fisica, junto con otras variables que existen en la base de datos. Ademas

utilice otras variables relevantes de la base de datos para explicar los factores latentes. Las variables a incluir en el modelo final deben tener sustento teorico y el modelo final debe optimizar el ajuste a los datos, en base a los criterios vistos en clase. Que puede concluir en base a sus resultados?

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.827
Number of iterations: 39
Params: 0.853 0.996 1.749 1.663 1.841 1.417 1.265 1.132 1.334 0.033 0.157 -0.017
-0.043 -0.015 -0.008 -0.341 0.087 -0.250 0.199 0.052 0.146 1.105 0.097 0.599
0.213 0.447 0.723 0.268 0.197 0.158 0.121 0.400 0.502 0.297
       DoF  DoF Baseline          chi2  chi2 p-value  chi2 Baseline       CFI  \
Value  156           177  32998.365517           0.0  139186.017356  0.763739

            GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.762919  0.731004  0.762919  0.731935  0.072641  66.345864

              BIC    LogLik
Value  358.544634  0.827068
          lval  op        rval  Estimate  Std. Err     z-value   p-value
0           F1   ~        sexo  0.032689  0.005357    6.101556       0.0
1           F1   ~        edad  0.157112  0.008146   19.286782       0.0
2           F2   ~  vive_padre -0.017199  0.002932   -5.865196       0.0
3           F2   ~  vive_madre -0.043206  0.008026   -5.383469       0.0
4           F3   ~       educm -0.015287  0.000836   -18.28351       0.0
5           F3   ~       educp -0.008323  0.000809  -10.285626       0.0
6          sk5   ~          F1  1.000000         -           -         -
7          sk6   ~          F1  0.853348  0.013956   61.147687       0.0
8          sk8   ~          F1  0.996151    0.0162   61.492288       0.0
9          sk1   ~          F2  1.000000         -           -         -
10         sk2   ~          F2  1.748600  0.020485   85.359582       0.0
11         sk3   ~          F2  1.662960  0.018826   88.331287       0.0
12         sk4   ~          F2  1.841407  0.020112   91.558782       0.0
13         sk9   ~          F3  1.000000         -           -         -
14        sk10   ~          F3  1.417024  0.017269    82.05442       0.0
15        sk11   ~          F3  1.264837  0.013916   90.890854       0.0
16        sk12   ~          F3  1.132025  0.014274   79.305688       0.0
17        sk13   ~          F3  1.333786  0.017456   76.408083       0.0
18  act_fisica   ~          F1 -0.340944  0.014469  -23.563489       0.0
19  act_fisica   ~          F2  0.086748  0.026817    3.234791  0.001217
20  act_fisica   ~          F3 -0.250368  0.015831  -15.814766       0.0
21          F1  ~~          F1  0.198646  0.003629   54.744576       0.0
22          F2  ~~          F2  0.051812  0.000937   55.282085       0.0
23          F3  ~~          F3  0.145988  0.002695   54.160035       0.0
24  act_fisica  ~~  act_fisica  1.104741  0.007914  139.589836       0.0
```

25	sk1	~~	sk1	0.096574	0.000844	114.481202	0.0
26	sk10	~~	sk10	0.599400	0.005149	116.401274	0.0
27	sk11	~~	sk11	0.213401	0.002485	85.883893	0.0
28	sk12	~~	sk12	0.446663	0.003719	120.101447	0.0
29	sk13	~~	sk13	0.722588	0.005865	123.195781	0.0
30	sk2	~~	sk2	0.268392	0.002404	111.649225	0.0
31	sk3	~~	sk3	0.196989	0.001889	104.308944	0.0
32	sk4	~~	sk4	0.158484	0.001856	85.384294	0.0
33	sk5	~~	sk5	0.120723	0.00311	38.819517	0.0
34	sk6	~~	sk6	0.399588	0.003598	111.070743	0.0
35	sk8	~~	sk8	0.502215	0.004664	107.668514	0.0
36	sk9	~~	sk9	0.297429	0.002557	116.306746	0.0

7. Repita el analisis de la Pregunta 6 usando para cada submuestra generada a partir de las clases latentes obtenida en la pregunta 5. Reporte diferencias entre los parametros entre los distintos grupos (si existieran). Hay alguna ganancia de estimar el modelo por grupos versus el modelo general?